
Linear Probes on Curved Latent Spaces: Probe-Facing Curvature Predicts Local Readout Accuracy

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Abstract

1 Linear probes are the standard way to read physical quantities out of foundation-
2 model embeddings, yet the embedding manifold is curved and the probe is not.
3 We ask which curvature a linear readout pays for and where on the manifold it can
4 be trusted. Restricted to the manifold, a linear function has Hessian $\langle w_N, \Pi \rangle$, the
5 second fundamental form contracted with the probe’s normal component; we turn
6 this classical identity into a per-point measurement on galaxy embeddings from
7 five encoders, using a decoder-based estimate of Π validated on a synthetic sur-
8 face of known curvature. Compared with the label’s Hessian along the manifold
9 (how the physical quantity itself curves over the embedding), the probe-facing cur-
10 vature predicts local readout accuracy: accuracy is higher where the two agree and
11 lower where they disagree, robustly under cross-fitting and across encoders, and a
12 counterfactual that reverses the probe-facing shape term at a fixed manifold lowers
13 local accuracy at nearly every anchor while a random orientation of matched
14 strength does not reproduce the gain. Mean curvature, the usual scalar summary,
15 keeps only the trace of Π and does not in general determine this; as an application,
16 a pre-registered mean-curvature statistic shows no density-controlled association
17 with cross-survey neighbour agreement. The mismatch tensor is a per-object diag-
18 nostic of where a linear readout of a given physical quantity is locally reliable, and
19 of which quantities an embedding carries linearly at all.

20 1 Introduction

21 Foundation-model embeddings of galaxy images are read out with linear probes: a ridge regression
22 from the embedding to a catalogue quantity such as magnitude, redshift or stellar mass [Smith et al.,
23 2024, Duraphe et al., 2025]. A probe returns a number for every object but no statement of where on
24 the embedding manifold that number can be trusted, and the manifold is curved while the probe is not.
25 We ask which curvature a linear readout pays for, and find it is a specific tensor whose agreement
26 with the curvature of the physical quantity itself decides local accuracy. That makes the disagreement
27 between the two a per-object diagnostic of readout reliability and, across quantities, an indication of
28 which physical properties an embedding carries linearly.

29 We contribute (i) a decoder-based pointwise mean-curvature and second-fundamental-form instru-
30 ment, validated on a synthetic surface of known curvature; (ii) the classical identity $\text{Hess}_M(w \cdot x) =$
31 $\langle w_N, \Pi \rangle$ applied to a fitted probe on learned manifolds; (iii) evidence on 86,471 labelled galaxies
32 and five encoders that the tensor mismatch between probe-facing curvature and the label’s Hessian
33 predicts local readout accuracy, and an intervention on that tensor with a matched control that tests
34 the mechanism; and (iv) as an application, a pre-registered density-controlled test in which mean
35 curvature has no robust association with cross-survey representational convergence.

2 Pointwise curvature from a decoder

Data. The readout experiments use 86,471 single-survey galaxies embedded by a supervised ImageNet-21k ViT-B/16 (“ViT-B” below; not DINOv3) with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]), and the same galaxies under four further encoders (Section 4). The application of Section 5 uses DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019] (Platonic Universe release [Duraphe et al., 2025]); we draw 10,000 rows once with a recorded seed. We unit-normalize embeddings (data on S^{767}).

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; we evaluate curvature on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$ (unnormalized trace; some authors divide by d), by automatic differentiation (cf. [Lee and Park, 2023, Acosta et al., 2023]), at points where DF has full column rank d (likewise $D(F/\|F\|)$ after projection). On the unit sphere the Euclidean mean-curvature vector of every d -submanifold has the universal radial component $-d\hat{x}$, which says nothing about shape; differentiating $F/\|F\|$ instead splits H exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}), and H^S carries the non-radial mean-curvature component. Sections 3–4 use the projected decoder; the application of Section 5 uses the pre-registered $\|H_{\text{raw}}\|$.

Validation. Against closed-form curvature, at $D = 768$ an in-sphere surface $G(z) = Q \text{normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h four Gaussian bumps, Q a rotation) gives rank 0.94 against the true $\|H^S\|$, direction cosine 0.999, magnitude ratio 1.00.

3 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$ [Lee, 2018], so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

For $M \subset S^{D-1}$, $\Pi = \Pi^S - g\hat{x}$ splits off the sphere’s radial direction, so $\text{Hess}_M(w \cdot x) = \langle w_S, \Pi^S \rangle - (w \cdot \hat{x})g$ (2), with w_S the part of w_N orthogonal to $\hat{x}$: the sphere bends every readout of a unit-normalized embedding in proportion to its own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 , with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$ (all tensor norms and cosines use $\langle A, B \rangle_g = g^{ia}g^{jb}A_{ij}B_{ab}$),

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2\|\delta\|^2 + \frac{1}{2}s^4\|\Delta\|_g^2 + O(s^4\|\delta\|) + O(s^6). \quad (3)$$

(The same expansion for a linear map between two embeddings gives a connection-mismatch and a relative extrinsic term; Appendix C.)

So $\|H\|$ need not correlate with $\langle w_N, \Pi \rangle$; and since the second-order term scales as s^4 with s set by density at fixed k , patch-scored curvature effects are density-sensitive.

4 Probe-facing curvature: bending toward the label helps, bending away hurts

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman–Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at five scales.

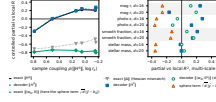


Figure 1: (a) Known surface, linear label, exact geometry: three-control partial vs local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder’s $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_S, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ _g$		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control. * not significant at 0.05.

Known surface. Resampling the validation surface with weight $\|H^S\|^\gamma$ (Figure 1a) moves the $\|H^S\|$ partial from -0.29 to $+0.26$ while $\|\langle w_N, \Pi \rangle\|_g$ holds at -0.74 to -0.80 , dominated by the sphere term of Eq. (2).

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99). We estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours (decoder-based Monge coordinates, so the finite-patch coefficient estimates the covariant Hessian). Three regularities follow (Table 1; Appendix A); the sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass. For magnitude, redshift and smooth fraction the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$); stellar mass shows no mismatch or alignment signal here. The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift): with $K = \langle w_N, \Pi \rangle$ the cross term of $\|\Delta\|_g^2 = \|\text{Hess}_M y\|_g^2 - 2\langle \text{Hess}_M y, K \rangle_g + \|K\|_g^2$ decides its sign. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55–0.65).

Bending toward helps, bending away hurts. Within the quadratic surrogate of Eq. (3), the Hessian-mismatch term is lower than at $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle_g > \|K\|_g^2$: the manifold must bend toward the label’s own curvature, and not too far. (The identity is classical, and Liu et al. [2023] already derive, for least squares on a hypersurface, a normal weight proportional to the label’s second derivatives over curvature, the low-dimensional special case of this mechanism; curved representations distorting a linear decoder is documented [Jurewicz et al., 2024], $\text{Hess}_M y$ is a classical object [Donoho and Grimes, 2003], and manifold capacity [Chung et al., 2016, Slatton et al., 2026], flattening [Psenka et al., 2024], curvature regularization [Ma et al., 2023], latent-curvature studies [Kaufman and Azencot, 2023], local-regression bias [Cheng and Wu, 2013] and geometry markers of generalization [Chou et al., 2026] use dimension, radius or magnitude. New here is the estimated pointwise tensor K on a learned manifold, its tensor comparison with $\text{Hess}_M y$, and the intervention that follows.) No anchor is flat, so we test the condition with a counterfactual local second-order surrogate of the readout. Writing $K = K_S - (w \cdot \hat{x})g$ with $K_S = \langle w_S, \Pi^S \rangle$ the in-sphere shape term, each anchor’s 2,048 neighbours are scored by $\hat{y}_t(u) = c + a^\top u + \frac{1}{2}K_{\text{sph}}(u, u) + \frac{t}{2}K_S(u, u)$ with a the readout’s tangent gradient, K_S from the decoder and c refit: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction (“bending toward the label” means $\langle R, K_S \rangle > 0$ for the remaining label curvature $R = \text{Hess}_M y - K_{\text{sph}}$), $t = -1$ the sign reversal of that probe-facing shape quadratic, $t = 0$ shape-flat. In the Gaussian model $t = 1$ beats shape-flat iff $2\langle R, K_S \rangle_g > \|K_S\|_g^2$; on the actual neighbours, with r_0 the centred shape-flat residual and q_c the centred quadratic $\frac{1}{2}K_S(u_i, u_i)$, the ex-

act criterion is $2\langle r_0, q_c \rangle > \|q_c\|^2$ with optimum $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ (Appendix E). On five encoders and all four labels, $t = 1$ beats shape-flat at 79–99% of anchors and $t = -1$ is worse at 97–100%, while a random orientation with the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign ($t = 1$ beats shape-flat at 22–36% of anchors, $t^* \approx 0$; Appendix E). The probe-selected shape quadratic is locally error-reducing; under the second-order model that is the signature of bending toward the remaining label curvature. This within-anchor question differs from Table 1’s cross-anchor one (does variation in mismatch explain variation in local R^2) and can hold where that has no signal, as for stellar mass. t^* sits at 1.5 to 3.6 throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial second-order term.

Across encoders. Across five encoders (the readout set’s supervised ViT-B, DINOv3 ViT-B/16, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$) the quadratic-chart mean-bending trace has density-controlled associations with the magnitude probe’s local R^2 of both signs (-0.24 to $+0.27$), as Eq. (1) allows. Per encoder (Appendix D) the decoder test is stable where the expansion predicts: the mismatch partial is negative for magnitude and redshift on all five encoders (-0.32 to -0.61 ; cross-fitted -0.28 to -0.62) and the alignment positive for magnitude on all five ($+0.13$ to $+0.53$) and for redshift on four; the shape term’s sign varies with encoder as well as label; for the mismatch and alignment hypotheses stellar mass stays null but for one mismatch cell (-0.12).

5 Application: mean curvature and cross-survey convergence

The Platonic Universe [Duraphe et al., 2025] asks whether different models see the same sky [Huh et al., 2024], scoring per-galaxy agreement between HSC and Legacy Survey embeddings by the mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$. If models disagreed most where the surface bends sharply, $\|H_{\text{raw}}\|$ should associate negatively with local agreement. The raw association is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$. But k -NN statistics are density-sensitive [Radovanović et al., 2010], and local density ($k = 30$ estimate on the Legacy embedding) correlates with both agreement and $\|H_{\text{raw}}\|$: residualizing on it leaves -0.024 , -0.066 and -0.022 . Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$), and subtracting the radial projection of H_{raw} (a projection, not a recomputation on $F/\|F\|$) collapses it to -0.023 .

6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold, seen in the readout’s normal direction, bends toward the label’s curvature and loses it where that bending is reversed: gain at a large majority of anchors and loss at nearly every anchor, on five encoders. The residual expansion names the quantity, and a decoder’s pointwise second fundamental form measures it at production scale, validated against a synthetic surface of known curvature for H^S and in substantial agreement with the data. Two consequences for practice. First, the mismatch tensor $\|\text{Hess}_M y - \langle w_N, \Pi \rangle\|_g$ is a per-object diagnostic: it says where a probe’s prediction of a given physical quantity rests on geometry the readout can follow, the ingredient a local reliability estimate for probe-based inference needs. Second, across quantities it separates labels the embedding carries linearly (magnitude, redshift, smooth fraction) from those it does not (stellar mass). Mean curvature, the scalar usually reported, keeps only the trace of Π and fails accordingly: it explains neither readout accuracy nor, in the application, cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate; its norm is reliably ranked (split-half ρ 0.91–0.94) but its direction only moderately (tensor cosine 0.19–0.35). On a synthetic surface with known curvature the mismatch partial reproduces but the alignment partial has no stable sign; Appendix E carries the alignment claim. The reliability reading is a diagnostic, not a calibrated interval. Mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

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221 Appendix A: decoder ablations

222 Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two
 223 further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach
 variance explained 0.952). Every significant sign of the main table is retained.

label	quantity	seed 0	seed 1	seed 2	400^3
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	−0.26	−0.26	−0.27	−0.26
	mismatch	−0.39	−0.35	−0.34	−0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	−0.06*	−0.09*	−0.09	−0.06*
	sphere	−0.33	−0.35	−0.34	−0.35
	mismatch	−0.45	−0.43	−0.42	−0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	−0.13	−0.15	−0.14	−0.11
	mismatch	−0.09*	−0.05*	−0.08*	−0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	−0.06*	+0.05*	+0.09
	sphere	−0.35	−0.35	−0.35	−0.35
	mismatch	−0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	−0.02*	+0.03*	−0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

224

225 Appendix B: sensitivity of the mismatch and sphere columns

226 Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local
 227 R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no
 228 rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak
 229 ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which
 230 substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that
 probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50	sphere, $\alpha = 1$ (main: $\alpha = 100$)
		fit A/score B	fit B/score A	fit A/score B	fit B/score A		
mag_r	16	−0.38	−0.34	+0.33	+0.31	+0.35	−0.21 (−0.26)
mag_r	20	−0.43	−0.40	+0.39	+0.37	+0.24	−0.22 (−0.26)
photo_z	16	−0.42	−0.36	+0.23	+0.20	+0.32	+0.05* (−0.33)
photo_z	20	−0.42	−0.42	+0.22	+0.22	+0.20	+0.03* (−0.36)
smooth_fraction	16	−0.10	−0.17	+0.17	+0.19	+0.33	−0.12 (−0.13)
smooth_fraction	20	−0.15	−0.21	+0.05*	+0.03*	+0.23	−0.13 (−0.15)
stellar_mass	16	+0.01*	−0.04*	−0.03*	+0.06*	+0.31	−0.02* (−0.35)
stellar_mass	20	−0.06*	−0.07*	+0.01*	−0.00*	+0.19	−0.02* (−0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

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Appendix C: relative second fundamental form, pilot

Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ; median first-order obstruction $\|AJ_F - J_GL\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of each decoder's $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$ (because first-order matching $AJ_F = J_GL$ fails at the 44% level, the second derivative of the alignment residual is not coordinate invariant; in the decoder chart its target-normal quadratic coefficient additionally contains $-P_N^G AJ_F \Gamma^F$, so relative II is reported only as an extrinsic diagnostic), the same with L fitted on 256 neighbours' latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F 's tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	-0.15	+0.04*	-0.18	+0.04*	-0.18	+0.13
$d = 25$, seed 0	0.64	0.43	-0.10	+0.06	-0.16	+0.05	-0.23	+0.13
$d = 20$, seed 1	0.64	0.44	-0.11	+0.05	-0.19	+0.04	-0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$, multi-scale density control. The first row per label is the probe's global out-of-sample R^2 ; Table 6 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per encoder.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	-0.11	+0.16	+0.16	-0.21
	sphere	-0.26	-0.23	-0.40	-0.24	-0.08*
	mismatch	-0.39	-0.57	-0.34	-0.32	-0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	-0.06*	-0.05*	-0.17	+0.29	-0.16
	sphere	-0.33	-0.21	-0.29	-0.46	-0.44
	mismatch	-0.45	-0.61	-0.45	-0.48	-0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	-0.08*	+0.20	+0.24	+0.06*
	sphere	-0.13	-0.06*	+0.05*	-0.15	-0.11
	mismatch	-0.09*	-0.33	-0.25	-0.20	-0.01*
	alignment	+0.19	-0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	-0.15	-0.10	-0.11	-0.13
	sphere	-0.35	-0.26	-0.38	-0.40	-0.38
	mismatch	-0.04*	+0.03*	-0.12	-0.06*	-0.02*
	alignment	+0.01*	-0.05*	-0.01*	+0.07*	-0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

Appendix E: intervening on the readout’s normal component

The partials of Table 1 compare anchors with one another and cannot say whether reversing the probe-facing shape term hurts relative to removing it, since no anchor is flat. Here we score a counterfactual local second-order surrogate of the readout at a fixed manifold. At each anchor the fitted ridge weight is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$; we score the anchor’s 2,048 neighbours by the data-side tangent-plus-sphere readout $(w_T + w_{\text{rad}}) \cdot x$ plus tq with the local intercept refit, where $q = \frac{1}{2} \langle w_S, \Pi^S \rangle(u, u)$ is the decoder’s in-sphere second-order term in the anchor’s chart coordinates: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction, $t = -1$ its sign reversal, $t = 0$ shape-flat (the sphere term $-(w \cdot \hat{x})g$ is kept in the base). Because the intercept is refit for every t , the local sum of squares is exactly $\text{SSE}(t) = \|r_0 - tq_c\|^2$ with r_0 the centred residual of the shape-flat predictor and $q_c = q - \bar{q}\mathbf{1}$ the centred quadratic on the anchor’s neighbours; hence $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ and $t = 1$ beats shape-flat iff $2\langle r_0, q_c \rangle > \|q_c\|^2$. This is the finite-sample counterpart of the tensor-level condition $2\langle R, K_S \rangle_g > \|K_S\|_g^2$ of Section 5, not the same quantity: the tensor version needs the estimated label Hessian, the empirical one does not. Columns: fraction of anchors where $t = 1$ beats shape-flat (help) and where $t = -1$ is worse than shape-flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ; then the same for a random in-sphere normal direction v whose quadratic is rescaled to the same centred amplitude on the actual neighbours, $\|q_{v,c}\| = \|q_c\|$ (matching $\|v\|$ to $\|w_S\|$ instead leaves the contracted tensor at a fraction of the fitted one, since Π^S spans at most $d(d+1)/2$ of the ~ 750 normal directions; matching $\|\langle v, \Pi^S \rangle\|_g$ gives the same picture; Supplement 12). The anchors’ neighbourhoods overlap (each point lies in about twelve of them), so the fractions are not counts of independent trials; a maximal set of anchors whose pairwise overlap is at most 5% of k has 19–21 members per run, on which $t = 1$ beats shape-flat at 81–100% (one-sided sign test $p \leq 4 \times 10^{-3}$ in every cell) and $t = -1$ is worse at 95–100% ($p \leq 3 \times 10^{-5}$).

run	label	decoder in-sphere shape term					random orientation, matched $\ q_c\ $				
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	-0.051	2.3	0.26	0.80	-0.009	-0.009	
	photo_z	0.99	0.99	+0.028	-0.038	3.0	0.33	0.68	-0.005	-0.006	
	smooth_fraction	0.90	1.00	+0.021	-0.040	1.7	0.24	0.78	-0.008	-0.010	
	stellar_mass	0.96	0.99	+0.026	-0.040	2.6	0.29	0.70	-0.006	-0.006	
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	-0.049	2.6	0.24	0.78	-0.007	-0.008	
	photo_z	0.99	1.00	+0.029	-0.038	3.6	0.36	0.71	-0.004	-0.005	
	smooth_fraction	0.90	1.00	+0.020	-0.035	1.9	0.22	0.77	-0.007	-0.008	
	stellar_mass	0.97	1.00	+0.027	-0.038	3.1	0.32	0.72	-0.005	-0.005	
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	-0.031	2.0	0.29	0.71	-0.007	-0.006	
	photo_z	0.96	0.99	+0.020	-0.030	2.6	0.29	0.71	-0.006	-0.004	
	smooth_fraction	0.93	1.00	+0.024	-0.055	1.5	0.22	0.78	-0.014	-0.013	
	stellar_mass	0.95	1.00	+0.016	-0.028	2.1	0.26	0.74	-0.005	-0.005	
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	-0.031	2.2	0.30	0.68	-0.006	-0.005	
	photo_z	0.99	1.00	+0.019	-0.026	3.4	0.36	0.64	-0.003	-0.003	
	smooth_fraction	0.92	0.99	+0.014	-0.027	1.7	0.25	0.69	-0.006	-0.005	
	stellar_mass	0.97	1.00	+0.020	-0.028	2.8	0.35	0.68	-0.004	-0.005	
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	-0.031	2.0	0.26	0.73	-0.007	-0.007	
	photo_z	0.99	1.00	+0.018	-0.026	2.9	0.28	0.64	-0.004	-0.003	
	smooth_fraction	0.88	0.98	+0.016	-0.036	1.6	0.23	0.75	-0.009	-0.008	
	stellar_mass	0.99	1.00	+0.022	-0.032	3.1	0.29	0.65	-0.005	-0.003	
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	-0.034	2.6	0.32	0.74	-0.005	-0.006	
	photo_z	0.97	0.99	+0.018	-0.025	3.2	0.34	0.67	-0.004	-0.003	
	smooth_fraction	0.79	0.97	+0.013	-0.029	1.6	0.25	0.77	-0.006	-0.008	
	stellar_mass	0.98	1.00	+0.022	-0.032	2.9	0.30	0.67	-0.004	-0.004	

Table 7: Counterfactual scaling of the in-sphere shape term in a local second-order surrogate of the readout, 512 anchors per run. The manifold’s bending as seen in the readout’s normal direction helps at a large majority of anchors, its sign reversal hurts at nearly every anchor, a random orientation of the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign; $t^* > 1$ throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial term.